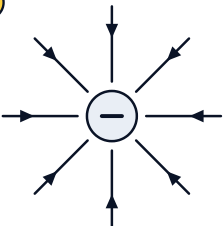
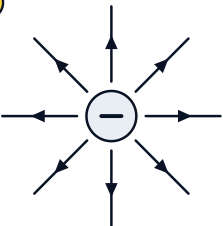


# Static charge and electric fields

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

|    | QUESTION                                                                                             | ANSWER                                                                          | ✓                        |
|----|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------|
| 1  | Describe how insulating materials can become electrically charged.                                   | By rubbing them together: friction transfers electrons between them.            | <input type="checkbox"/> |
| 2  | Describe how a material can become negatively charged.                                               | By gaining electrons.                                                           | <input type="checkbox"/> |
| 3  | Describe how a material can become positively charged.                                               | By losing electrons.                                                            | <input type="checkbox"/> |
| 4  | Describe what happens when two charged objects are brought close together.                           | They exert a force on each other: like charges repel, opposite charges attract. | <input type="checkbox"/> |
| 5  | State whether electrostatic force is a contact force or non-contact force.                           | Non-contact force                                                               | <input type="checkbox"/> |
| 6  | Explain why there is no potential difference between two insulators before they are rubbed together. | Both are neutral, so there is no difference in charge.                          | <input type="checkbox"/> |
| 7  | Name the type of field that surrounds every charged object.                                          | An electric field                                                               | <input type="checkbox"/> |
| 8  | State where the electric field around a charged object is strongest.                                 | Closest to the object.                                                          | <input type="checkbox"/> |
| 9  | Describe how the strength of an electric field changes with distance from the charge.                | The field gets stronger closer to the charge.                                   | <input type="checkbox"/> |
| 10 | State the direction in which electric field lines point.                                             | Away from positive charges and towards negative charges.                        | <input type="checkbox"/> |

| QUESTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ANSWER                                                                                                         | ✓                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------|
| <p>11 Which diagram, A or B, shows the electric field around a negative charge? Explain.</p> <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;"> <p><b>A</b></p>  </div> <div style="text-align: center;"> <p><b>B</b></p>  </div> </div> | <p>A.<br/>The field lines point towards the negative charge.</p>                                               | <input type="checkbox"/> |
| <p>12 Explain why a charged object experiences a force when close to another charged object.</p>                                                                                                                                                                                                                                                                                                                                                                           | <p>The electric field of each object exerts a force on the other object.</p>                                   | <input type="checkbox"/> |
| <p>13 State how the size of the force between two charged objects can be increased.</p>                                                                                                                                                                                                                                                                                                                                                                                    | <p>By moving them closer together.</p>                                                                         | <input type="checkbox"/> |
| <p>14 State what happens when two objects with the same type of charge are brought together.</p>                                                                                                                                                                                                                                                                                                                                                                           | <p>They repel each other.</p>                                                                                  | <input type="checkbox"/> |
| <p>15 State what happens when two objects with opposite charges are brought together.</p>                                                                                                                                                                                                                                                                                                                                                                                  | <p>They attract each other.</p>                                                                                | <input type="checkbox"/> |
| <p>16 A plastic rod becomes negatively charged when it is rubbed with a cloth. State and explain the charge on the cloth.</p>                                                                                                                                                                                                                                                                                                                                              | <p>Positive (equal in size): the cloth lost the electrons that the rod gained.</p>                             | <input type="checkbox"/> |
| <p>17 What is an electric field?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>The region around a charged object where another charged object experiences a force.</p>                    | <input type="checkbox"/> |
| <p>18 Describe the electric field around an isolated positively charged sphere.</p>                                                                                                                                                                                                                                                                                                                                                                                        | <p>Straight lines pointing radially outwards, away from the sphere.</p>                                        | <input type="checkbox"/> |
| <p>19 Explain how a spark is produced between a charged object and an earthed conductor.</p>                                                                                                                                                                                                                                                                                                                                                                               | <p>The strong electric field ionises the air, which then conducts: charge flows across the gap as a spark.</p> | <input type="checkbox"/> |